Name: Trinh Gia Hiep - 2352343 - HW2

1) Ranking from low → high impurity:
(12,0) < (9,3) < (6,6)

- Intuition: Entropy measures uncertainty about a class of a sample.
(12,0): Every sample is in the same class.
→ Minimum impurity, no uncertainty
(9,3): 75/25 split → Some uncertainty, majority is right.
(6,6): 50/50 split → Maximum impurity.

- $H = -\sum p_i \log_2 p_i$:
(12,0) → $H = 0$
(9,3) → $H = -(0,75 \cdot \log_2 0,75 + 0,25 \cdot \log_2 0,25) = 0,811$
(6,6) → $H = -(2 \cdot \log_2 0,5) = 1$

2) Parent node: (11P, 7N) ⇒ N = 18
$Gini = 1 - \sum p_i^2 = 1 - (11/18)^2 - (7/18)^2 = 0,4753$

- S1: (6P, 1N), (5P, 6N)

(6P, 1N): $1 - \frac{36+1}{49} = 0,2449$

(5P, 6N): $1 - \frac{25+36}{121} = 0,4959$

Weighted $= \frac{7}{18} 0,2449 + \frac{11}{18} 0,4959$
$= 0,3983$

Gain $= 0,4753 - 0,3983$
$= 0,077$

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- S2: (8P, 4N) & (3P, 3N)

(8P, 4N): $1 - \frac{8^2+4^2}{12^2} = 0,444$

(3P, 3N): $1 - \frac{9+9}{36} = 0,5$

Weighted $= \frac{12}{18} 0,444 + \frac{6}{18} 0,5$
$= 0,463$

Gain $= 0,4753 - 0,463 = 0,0123$

⇒ CART prefers S1 because it has the lower weighted Gini.
- CART scores the split only by how pure the children are, not equal child sizes.

3. Data x = 4, 7, 9, 13, 18, 21, 29 with labels:
N, N, N, P, P, N, P (3P, 4N)

Parent Gini $= 1 - \frac{9+16}{49} = 0,4898$

- Cut $x \le 8$: Left {4, 7} Left (P,N) = (0,2) → Gini = 0
Right (P,N) = (3,2) → Gini $= 1 - \frac{9+4}{25} = 0,48$

⇒ Gini_LR = 0 / 0,48 ⇒ Weighted Gini $= \frac{2}{7} \cdot 0 + \frac{5}{7} \cdot 0,48 = 0,343$

Cut	Left (P,N)	Right (P,N)	Gini_L	Gini_R	Weighted Gini	Gain
$x \le 8$	(0,2)	(3,2)	0	0,48	$(5/7) \cdot 0,48 = 0,343$	0,148
$x \le 11$	(0,3)	(3,1)	0	0,375	$(4/7) \cdot 0,375 = 0,214$	0,276
$x \le 19,5$	(2,3)	(1,1)	0,48	0,5	$(5/7) \cdot 0,48 + (2/7) \cdot 0,5 = 0,486$	0,004

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⇒ $x \leq 11$ is clearly best because it isolates a pure left child, while $x \leq 19,5$ barely reduces any impurity.

4. Pruning : The ~~left~~ leaf makes 6 errors, ~~not~~ as worse as the subtree's 7. Under reduced-error pruning, we prune and replace the subtree with a leaf, since a simpler model generalizes at least as well.

- Missing values :

a) Surrogate splits (CART) : Route the sample using the feature that best mimics the primary split.

⇒ Drawback : If no other feature is strongly correlated with the primary, it ~~not~~ will work poorly.

b) Fractional weighting (C4.5) : Send the sample down all branches, weighted by observed branch frequencies.

⇒ Drawback : Inference is more expensive, it assumes the values are missing at random.

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	Baseline	max_depth = 5
Max depth	18	5
Leaves	163	17
Train score	1,000	~~0,880~~ 0,920
Validation score	0,8455	0,8431
Top 3 features	tGravityAcc-min()-X, fBodyAccJerk-bandsEnergy()-1,16 angle(X, gravityMean)	same

. Prediction. With max_depth = 5, depth = 5 and leaves ≤ 32 :

① Train score drops to 0,88 - 0,92

② Validation score stays the same or maybe rises a bit (0,84 - 0,85)

③ Train-validation gap thinks to 5 points

④ Top 3 features are the same.

- Comparison : ① Train score fell to 0,92

② Validation changed only -0,2 points.

③ The gap thrank from 15,5 to 7,7 points ≤ 5 points predicted.

④ Top 3 features kept their order.

My prediction assumed the deep branches were mostly noise. Therefore validation didn't rise.

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